

第2章 分离变量法

2.1 有界弦的自由振动

2.2 有限长杆上的热传导

2.3 圆域内二维拉普拉斯方程的定解问题

2.4 非齐次方程的解法

2.5 非齐次边界条件的处理

2.6 本征值问题



预备知识

常微分方程 傅里叶级数

二阶常系数线性常微分方程的标准形式

1. 齐次常微分方程

$$y'' + py' + qy = 0 \xrightarrow[\text{特征方程}]{\text{特征}} r^2 + pr + q = 0$$

特征根的情况	通解的表达式
实根 $r_1 \neq r_2$	$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
实根 $r_1 = r_2$	$y = (C_1 + C_2 x) e^{r_2 x}$
复根 $r_{1,2} = \alpha \pm i\beta$	$y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$

2. 非齐次常微分方程

$$y'' + py' + qy = f(x)$$

$$\xrightarrow{\text{对应齐次方程}} y'' + py' + qy = 0$$

通解结构 $y(x) = Y(x) + y^*(x)$

其中 $y^*(x)$ 是非齐次方程的一个特解,
 $Y(x)$ 是对应齐次方程的通解.

$f(x)$ 以 $2l$ 为周期的傅里叶级数

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x \right)$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi}{l} x \, dx, n = 0, 1, 2, \dots$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi}{l} x \, dx, n = 1, 2, \dots$$

奇函数 $f(x) \sim \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi}{l} x \, dx,$$
$$n = 1, 2, \dots$$

偶函数 $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{l} x$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi}{l} x \, dx,$$
$$n = 0, 1, 2, \dots$$

分离变量法

又称特征展开法、Fourier级数方法和驻波法，是求解偏微分方程最基本和常用的方法。

- **理论依据：**线性方程的叠加原理和施图姆-刘维尔 (Sturm-Liouville) 本征值理论。
- **基本思想：**将求解偏微分方程的定解问题设法转化为常微分方程(ODE)的定解问题。

本章层次：

齐次方程 + 齐次边界条件



非齐次方程 + 齐次边界条件



非齐次方程 + 非齐次边界条件



2.1 有界弦的自由振动

两端固定弦的自由振动的定解问题:

$$\frac{\partial^2 u(x,t)}{\partial t^2} = a^2 \frac{\partial^2 u(x,t)}{\partial x^2}, \quad 0 < x < l, \quad t > 0$$

$$\text{初始条件: } u(x,0) = \varphi(x), \quad \frac{\partial u(x,0)}{\partial t} = \psi(x), \quad 0 \leq x \leq l$$

$$\text{边界条件: } u(0,t) = u(l,t) = 0, \quad t > 0 \quad \boxed{\text{I/I类齐次边界条件}}$$

方程特点: 偏微分方程是线性齐次的, 边界条件是齐次的.

基本思想: 首先求出具有变量分离形式且满足边界条件的特解, 然后由叠加原理作出这些解的线性组合, 最后由其余的定解条件确定叠加系数.

第1步：分离变量，取试探解 $u(x,t) = X(x)T(t)$ ；
(可分离变量的先决条件是偏微分方程和边界条件都是齐次的)

将解 $u(x,t) = X(x)T(t)$ 代入波动方程，有

$$X(x)T''(t) = a^2 X''(x)T(t) \Rightarrow \frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)}$$

其比值只能是常数，令比值 $= -\lambda$ ，则：

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ T''(t) + \lambda a^2 T(t) = 0 \end{cases}$$

第2步：求本征值 λ 和本征函数 $X(x)$ ；

$$\text{代入边界条件} \begin{cases} u(0,t) = 0 \rightarrow X(0)T(t) = 0 \rightarrow X(0) = 0 \\ u(l,t) = 0 \rightarrow X(l)T(t) = 0 \rightarrow X(l) = 0 \end{cases}$$

转化为常微分方程的边值问题

$$X''(x) + \lambda X(x) = 0, \quad \text{边界条件: } X(0) = X(l) = 0$$

本征值 (特征值或固有值) 问题: 即含有待定常数的常微分方程在一定条件下求非零解的问题. 其中

本征值 (特征值, 固有值): 使方程有非零解的待定常数值
本征 (特征, 固有) 函数: 与本征值相对应的ODE非零解

针对 λ 分情况讨论:

1) 若 $\lambda < 0$, 则通解为 $X(x) = Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x}$

代入边界条件:
$$\begin{cases} A + B = 0 \\ Ae^{\sqrt{-\lambda}l} + Be^{-\sqrt{-\lambda}l} = 0 \end{cases} \Rightarrow A = B = 0 \Rightarrow X = 0$$

无意义

2) 若 $\lambda = 0$, 则通解为 $X(x) = Ax + B$

$$\text{代入边界条件: } \begin{cases} B = 0 \\ Al + B = 0 \end{cases} \Rightarrow A = B = 0 \Rightarrow X = 0$$

无意义

3) 若 $\lambda > 0$, 令 $\lambda = \beta^2$ (β 为非零实数),

则通解为 $X(x) = A \cos \beta x + B \sin \beta x$

$$\text{代入边界条件: } \begin{cases} A = 0 \\ A \cos \beta l + B \sin \beta l = 0 \end{cases} \Rightarrow \begin{cases} A = 0 \\ \sin \beta l = 0 \end{cases}$$

$$\Rightarrow \beta = \frac{n\pi}{l} \quad (n = 1, 2, \dots) \Rightarrow \lambda = \frac{n^2 \pi^2}{l^2} \quad (n = 1, 2, \dots)$$

$$\therefore \begin{cases} \text{本征值: } \lambda_n = \frac{n^2 \pi^2}{l^2}, \quad n = 1, 2, \dots \\ \text{本征函数: } X_n(x) = \sin \frac{n\pi}{l} x, \quad n = 1, 2, 3, \dots \end{cases}$$

本征函数的正交性:

$$\int_0^l X_n(x) X_m(x) dx = 0, \quad n \neq m$$

证明: $\because X''(x) + \lambda X(x) = 0$, 边界条件: $X(0) = X(l) = 0$

$$\therefore \begin{cases} X_n''(x) + \lambda_n X_n(x) = 0, & X_n(0) = X_n(l) = 0 \\ X_m''(x) + \lambda_m X_m(x) = 0, & X_m(0) = X_m(l) = 0 \end{cases}$$

用 $X_m(x)$ 乘以 $X_n(x)$ 的方程, 用 $X_n(x)$ 乘以 $X_m(x)$ 的方程, 相减并积分得:

$$\begin{aligned} (\lambda_n - \lambda_m) \int_0^l X_n(x) X_m(x) dx &= \int_0^l [X_n(x) X_m''(x) - X_m(x) X_n''(x)] dx \\ &= [X_n(x) X_m'(x) - X_m(x) X_n'(x)] \Big|_0^l = 0 \end{aligned}$$

结论: 对定解问题, 只要边界条件是齐次的, 则本征函数一定正交.

第3步: 求被分离的另一函数 $T(t)$;

将本征值代入 $T''(t) + \lambda a^2 T(t) = 0$ 有

$$T_n''(t) + \frac{n^2 \pi^2}{l^2} a^2 T_n(t) = 0, \quad n = 1, 2, \dots$$

$$T_n(t) = C_n' \cos \frac{n\pi a}{l} t + D_n' \sin \frac{n\pi a}{l} t \quad (n = 1, 2, 3, \dots)$$

第4步: 叠加特解求一般解;

$$u_n(x, t) = T_n(t) X_n(x) = \left(C_n \cos \frac{n\pi a}{l} t + D_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi}{l} x$$

这些特解满足方程和齐次边界条件, 但不满足初始条件. 为了表征初始条件, 由线性方程的叠加原理, 设原问题的解为:

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left(C_n \cos \frac{n\pi a}{l} t + D_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi}{l} x$$

第5步：确定常数

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left(C_n \cos \frac{n\pi a}{l} t + D_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi}{l} x$$

由初始条件：

$$u(x, t) \Big|_{t=0} = u(x, 0) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{l} x = \varphi(x)$$

$$\frac{\partial u(x, t)}{\partial t} \Big|_{t=0} = \sum_{n=1}^{\infty} D_n \frac{n\pi a}{l} \sin \frac{n\pi}{l} x = \psi(x), \quad x \in [0, l].$$

若将函数 $\varphi(x)$ 和 $\psi(x)$ 作奇延拓后再延拓为周期 $2l$ 的周期函数, 则 C_n 、 D_n 可看作两周期函数的傅里叶级数的系数.

$$\therefore C_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx, \quad D_n = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx$$

或由本征函数的正交性求系数:

$$\int_0^l X_n(x) X_m(x) dx = 0, \quad n \neq m$$

$$\int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx = \int_0^l \sum_{m=1}^{\infty} C_m \sin \frac{m\pi}{l} x \sin \frac{n\pi}{l} x dx = \frac{l}{2} C_n$$

$$\therefore C_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx$$

$$\int_0^l \psi(x) \sin \frac{n\pi}{l} x dx = \int_0^l \sum_{m=1}^{\infty} D_m \frac{m\pi a}{l} \sin \frac{m\pi}{l} x \sin \frac{n\pi}{l} x dx = \frac{n\pi a}{2} D_n$$

$$\therefore D_n = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx$$

$$u = \sum_{n=1}^{\infty} \left[\left(\frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx \right) \cos \frac{n\pi a}{l} t + \left(\frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx \right) \sin \frac{n\pi a}{l} t \right] \sin \frac{n\pi}{l} x$$

分离变量法概要：

- (1) 分离变量, 将偏微分方程化为两个常微分方程
- (2) 确定本征值和本征函数（利用边界条件）
- (3) 确定一般解（级数形式解）
- (4) 确定一般解中的待定常数（利用初始条件）

分离变量法步骤

1. 定解问题: $\frac{\partial^2 u(x,t)}{\partial t^2} = a^2 \frac{\partial^2 u(x,t)}{\partial x^2}, \quad 0 < x < l, \quad t > 0$

初始条件: $u(x,0) = \varphi(x), \frac{\partial u(x,0)}{\partial t} = \psi(x), \quad 0 \leq x \leq l$

边界条件: $u(0,t) = u(l,t) = 0, \quad t > 0$

(1) 分离变量 $u(x,t) = X(x)T(t) \Rightarrow X'' + \lambda X = 0, \quad T'' + \lambda a^2 T = 0$

(2) 求本征值和本征函数 $\lambda_n = n^2 \pi^2 / l^2, \quad X_n(x) = \sin \frac{n\pi}{l} x$

(3) 求另一个函数 $T_n = C'_n \cos \frac{n\pi a}{l} t + D'_n \sin \frac{n\pi a}{l} t$

(4) 求一般解 $u = \sum_{n=1}^{\infty} T_n X_n = \sum_{n=1}^{\infty} (C_n \cos \frac{n\pi a}{l} t + D_n \sin \frac{n\pi a}{l} t) \sin \frac{n\pi}{l} x$

(5) 确定常数 $C_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx, \quad D_n = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx$

分离变量法可以求解具有齐次边界条件的齐次偏微分方程。

2. 解的性质 (物理意义)

对任何一个特解:

$$\begin{aligned} u_n(x, t) &= (C_n \cos \frac{n\pi a}{l} t + D_n \sin \frac{n\pi a}{l} t) \sin \frac{n\pi}{l} x \\ &= A_n \cos(\omega_n t - \theta_n) \sin \frac{n\pi}{l} x \end{aligned}$$

$$\text{其中: } A_n = \sqrt{C_n^2 + D_n^2}, \quad \omega_n = \frac{n\pi a}{l}, \quad \theta_n = \arctan \frac{D_n}{C_n}.$$

(1) 某一时刻($t = t_0$)弦上各点的一个振动方程:

$$u_n(x, t_0) = A_n \cos(\omega_n t_0 - \theta_n) \sin \frac{n\pi}{l} x, \quad 0 < x < l$$

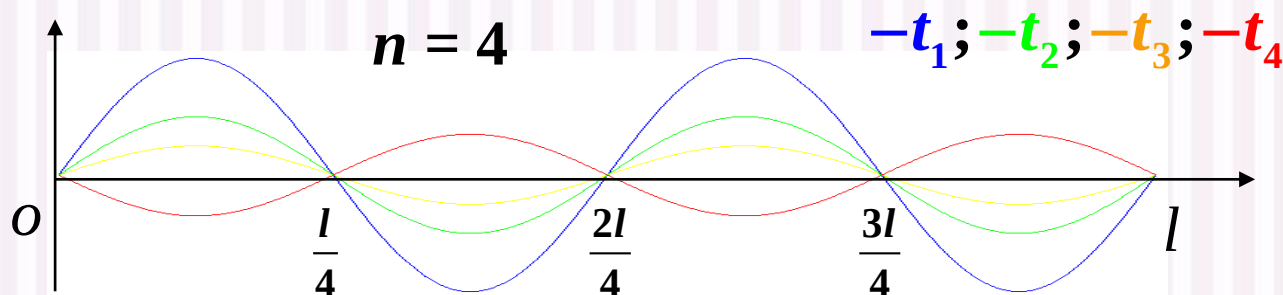
表明任意时刻, 波形的形状都是一些正弦曲线.

(2) 弦上某点 ($x = x_0$) 的一个振动方程:

$$u_n(x_0, t) = A_n \sin \frac{n\pi}{l} x_0 \cos(\omega_n t - \theta_n), \quad t \geq 0$$

★ 在 $x = \frac{ml}{n} (m = 0, 1, \dots, n)$ 各点上, 振动的振幅恒为0, 称为节点(波节), 这种包含节点的振动波叫做**驻波**;

★ 在 $x = \frac{(2m+1)l}{2n} (m = 0, 1, \dots, n-1)$ 各点上, 振动的振幅的绝对值恒为最大, 称为波峰(波腹).



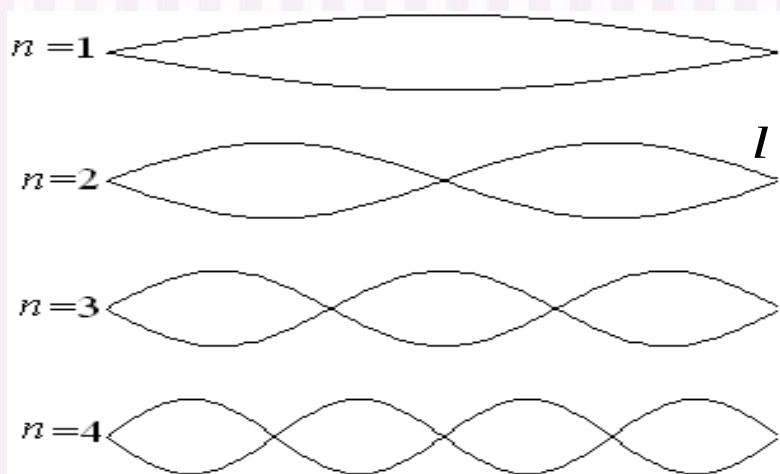
(3) 每个特解代表一个驻波 (standing wave):

$$u_n(x, t) = A_n \cos(\omega_n t - \theta_n) \sin \frac{n\pi}{l} x, \quad 0 < x < l$$

驻波波长: $\lambda_n = \frac{2}{n} l$ 驻波频率: $f_n = \frac{\omega_n}{2\pi} = \frac{na}{2l}$

驻波波速: $v = f_n \lambda_n = \frac{na}{2l} \cdot \frac{2l}{n} = a = \sqrt{\frac{T}{\rho}}$

驻
波
法



基频: $f_1 = \frac{a}{2l},$

倍频: $f_n = n \frac{a}{2l}, \quad n = 2, 3, \dots$

基频决定弦的音调, 当 ρ 一定时, 可调节弦的绷紧程度(张力 T) 以改变音调. 倍频决定音色.

例 求下列定解问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, & 0 < x < 1, t > 0 \\ u(0, t) = u(1, t) = 0, & t > 0 \\ u(x, 0) = \sin \pi x, \frac{\partial u(x, 0)}{\partial t} = 0, & 0 \leq x \leq 1 \end{cases}$$

解 Step1: 分离变量

令 $u(x, t) = X(x)T(t)$ 代入偏微分方程有:

$$XT'' = X''T$$

$$\frac{X''}{X} = \frac{T''}{T} = -\lambda \Rightarrow \begin{cases} X'' + \lambda X = 0 \\ T'' + \lambda T = 0 \end{cases}$$

——2.1 有界弦的自由振动——

考虑定解边界条件

$$\begin{aligned} u(0, t) = X(0)T(t) = 0 \\ u(1, t) = X(1)T(t) = 0 \end{aligned} \Rightarrow X(0) = 0, X(1) = 0$$

Step2: 解本征方程, 求本征值和本征函数

$$\begin{cases} X'' + \lambda X = 0, & 0 < x < 1 \\ X(0) = 0, X(1) = 0 \end{cases}$$

Case 1: $\lambda = -\beta^2 < 0$

$$X'' - \beta^2 X = 0 \rightarrow X(x) = Ae^{\beta x} + Be^{-\beta x}$$

$$\begin{cases} X(0) = A + B = 0 \\ X(1) = Ae^{\beta} + Be^{-\beta} = 0 \end{cases} \Rightarrow A = B = 0 \Rightarrow X(x) = 0$$

无意义

——2.1 有界弦的自由振动——

Case 2: $\lambda = 0$

$$X'' = 0 \Rightarrow X(x) = Ax + B \Rightarrow A = B = 0 \Rightarrow X(x) = 0$$

无意义

Case 3: $\lambda = \beta^2 > 0$

$$X'' + \beta^2 X = 0 \Rightarrow X(x) = A \cos \beta x + B \sin \beta x$$

$$\begin{cases} X(0) = A = 0 \\ X(1) = B \sin \beta = 0 \end{cases} \Rightarrow \begin{cases} A = 0 \\ \beta_n = n\pi, n = 1, 2, 3, \dots \end{cases}$$

$$\therefore \lambda_n = n^2 \pi^2, X_n(x) = B_n \sin n\pi x$$

Step3: 求另一函数

$$\begin{aligned} T'' + \lambda T &= 0 \\ \lambda_n &= n^2 \pi^2 \Rightarrow T_n'' + n^2 \pi^2 T_n = 0 \Rightarrow T_n = C'_n \cos n\pi t + D'_n \sin n\pi t \end{aligned}$$

Step4: 叠加求一般解

$$\begin{aligned}u_n &= X_n T_n = B_n \sin n\pi x (C'_n \cos n\pi t + D'_n \sin n\pi t) \\&= (C_n \cos n\pi t + D_n \sin n\pi t) \sin n\pi x\end{aligned}$$

$$u = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} (C_n \cos n\pi t + D_n \sin n\pi t) \sin n\pi x$$

Step5: 确定系数

$$u(x, 0) = \sum_{n=1}^{\infty} C_n \sin n\pi x = \sin \pi x \Rightarrow C_n = \begin{cases} 1, & n = 1 \\ 0, & n = 2, 3, \dots \end{cases}$$

$$\frac{\partial u(x, 0)}{\partial t} = \sum_{n=1}^{\infty} D_n n\pi \sin n\pi x = 0 \Rightarrow D_n = 0$$

$\therefore$ 原问题的解为 $u = \cos \pi t \cdot \sin \pi x, 0 < x < 1, t > 0$

两端自由的均匀杆的自由纵振动

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, & x \in (0, l), t > 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & x \in [0, l] \\ u_x(0, t) = u_x(l, t) = 0, & t \geq 0 \end{cases} \quad \text{II/II类齐次边界条件}$$

解：设 $u(x, t) = X(x)T(t)$

$$XT'' = a^2 X''T$$

$$\frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda$$

$$\begin{cases} \text{本征值问题} \begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases} \\ T''(t) + a^2 \lambda T(t) = 0 \end{cases}$$

——2.1 有界弦的自由振动——

(i) $\lambda < 0$ 时, $X(x) = C_1 e^{\sqrt{-\lambda}x} + C_2 e^{-\sqrt{-\lambda}x}$

$$\begin{cases} X'(0) = \sqrt{-\lambda}(C_1 - C_2) = 0 \\ X'(l) = \sqrt{-\lambda}(C_1 e^{\sqrt{-\lambda}l} - C_2 e^{-\sqrt{-\lambda}l}) = 0 \end{cases} \Rightarrow C_1 = C_2 = 0 \Rightarrow X \equiv 0$$

无意义

(ii) $\lambda = 0$ 时, $X(x) = C_0 + D_0 x$

$$\begin{cases} X'(0) = D_0 = 0 \\ X'(l) = D_0 = 0 \end{cases} \Rightarrow X(x) = C_0 \quad \longrightarrow$$

可看做是(iii)中
 $n = 0$ 的情形

(iii) $\lambda > 0$ 时, $X(x) = C_1 \cos \sqrt{\lambda}x + C_2 \sin \sqrt{\lambda}x$

$$\begin{cases} X'(0) = C_2 = 0 \\ X'(l) = C_1 \sqrt{\lambda} \sin \sqrt{\lambda}l = 0 \end{cases} \Rightarrow \sin \sqrt{\lambda}l = 0$$

$$\therefore \lambda = \frac{n^2 \pi^2}{l^2}, \quad X(x) = C_1 \cos \frac{n\pi x}{l}, \quad n = 1, 2, \dots$$

——2.1 有界弦的自由振动——

本征值 $\lambda = \frac{n^2 \pi^2}{l^2}, \quad n = 0, 1, 2, \dots$

本征函数 $X(x) = C_1 \cos \frac{n\pi x}{l}, \quad n = 0, 1, 2, \dots$

$$T''(t) + a^2 \lambda T(t) = 0 \Rightarrow \begin{cases} T_n'' = 0, & n = 0 \\ T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n = 0, & n \neq 0 \end{cases}$$

$$\therefore \begin{cases} T_0(t) = A_0 + B_0 t \\ T_n(t) = A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l}, \quad n = 1, 2, \dots \end{cases}$$

$$\therefore u(x, t) = A_0 + B_0 t + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l} \right) \cos \frac{n\pi x}{l}$$

——2.1 有界弦的自由振动——

$$u(x, t) = A_0 + B_0 t + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} \right) \cos \frac{n\pi x}{l}$$

$$\begin{cases} u(x, 0) = \varphi(x) \Rightarrow A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l} = \varphi(x) \\ u_t(x, 0) = \psi(x) \Rightarrow B_0 + \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \cos \frac{n\pi x}{l} = \psi(x) \end{cases}$$

$$\therefore \begin{cases} A_0 = \frac{1}{l} \int_0^l \varphi(x) dx \\ B_0 = \frac{1}{l} \int_0^l \psi(x) dx \end{cases} \quad \begin{cases} A_n = \frac{2}{l} \int_0^l \varphi(x) \cos \frac{n\pi x}{l} dx \\ B_n = \frac{2}{n\pi a} \int_0^l \psi(x) \cos \frac{n\pi x}{l} dx \end{cases}$$

混合边界条件的定解问题

■左端点固定、右端点自由的I/II类齐次边界条件

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, & x \in (0, l), t > 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & x \in [0, l] \\ u(0, t) = u_x(l, t) = 0, & t \geq 0 \end{cases} \quad \text{I/II类齐次边界条件}$$

本征值问题:

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$$

$\lambda \leq 0$ 时, $X(x) \equiv 0$, 无意义

$\lambda > 0$ 时, $X(x) = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x$

$$\begin{cases} X(0) = C_1 = 0 \\ X'(l) = C_2 \sqrt{\lambda} \cos \sqrt{\lambda} l = 0 \end{cases} \Rightarrow \cos \sqrt{\lambda} l = 0$$

$$\text{本征值: } \lambda_n = \left[\frac{\left(n + \frac{1}{2}\right)\pi}{l} \right]^2, \quad n = 0, 1, 2, \dots$$

$$\text{本征函数: } X_n(x) = C_2 \sin \left[\frac{\left(n + \frac{1}{2}\right)\pi}{l} x \right]$$

■左端点自由、右端点固定的II/I类齐次边界条件

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, & x \in (0, l), t > 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & x \in [0, l] \\ u_x(0, t) = u(l, t) = 0, & t \geq 0 \end{cases}$$

II/I类齐次边界条件

本征值问题:

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$$

$\lambda \leq 0$ 时, $X(x) \equiv 0$, 无意义

$\lambda > 0$ 时, $X(x) = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x$

$$\begin{cases} X'(0) = C_2 = 0 \\ X(l) = C_1 \cos \sqrt{\lambda} l = 0 \end{cases} \Rightarrow \cos \sqrt{\lambda} l = 0$$

$$\text{本征值: } \lambda_n = \left[\frac{\left(n + \frac{1}{2}\right)\pi}{l} \right]^2, \quad n = 0, 1, 2, \dots$$

$$\text{本征函数: } X_n(x) = C_1 \cos \left[\frac{\left(n + \frac{1}{2}\right)\pi}{l} x \right]$$



2.2 有限长杆上的热传导

杆的热传导问题（左端连等温热源，右端绝热）

$$\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < l, \quad t > 0$$

初始条件: $u(x, 0) = \varphi(x), \quad 0 \leq x \leq l$

I/III类齐次边界条件

边界条件: $u(0, t) = 0, \quad \frac{\partial u(l, t)}{\partial x} + hu(l, t) = 0, \quad t > 0$

第1步：分离变量，取试探解 $u(x, t) = X(x)T(t)$ ；

将解 $u(x, t) = X(x)T(t)$ 代入波动方程，有

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{a^2 T(t)} = -\lambda \Rightarrow \begin{cases} X''(x) + \lambda X(x) = 0 \\ T'(t) + \lambda a^2 T(t) = 0 \end{cases}$$

第2步：求本征值 λ 和本征函数 $X(x)$;

代入边界条件

$$\begin{cases} u(0,t) = 0 \rightarrow X(0)T(t) = 0 \rightarrow X(0) = 0 \\ \frac{\partial u(l,t)}{\partial x} + hu(l,t) = 0 \rightarrow X'(l) + hX(l) = 0 \end{cases}$$

转化为本征值问题

$$X''(x) + \lambda X(x) = 0 ,$$

$$\text{边界条件: } X(0) = 0, X'(l) + hX(l) = 0$$

针对 λ 分三种情况讨论可知, $\lambda \leq 0$ 时只有零解, 无意义, 因此仅考虑 $\lambda = \beta^2 > 0$ 的情形.

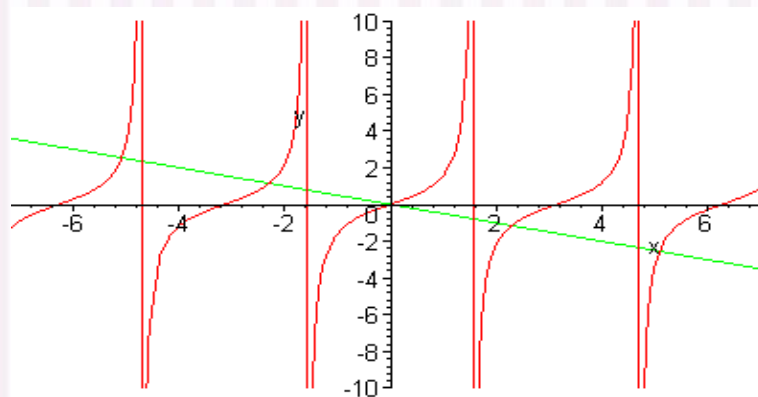
当 $\lambda = \beta^2 > 0$ 时 (β 为非零实数)

则通解为 $X(x) = A \cos \beta x + B \sin \beta x$

代入边界条件:
$$\begin{cases} X(0) = A = 0 \\ X'(l) + hX(l) = B\beta \cos \beta l + Bh \sin \beta l = 0 \end{cases}$$

$\Rightarrow \tan \beta l = -\frac{\beta}{h}$, β 可由正切函数

与直线函数的无穷个交点确定, 取正根 β_n , $n = 1, 2, \dots$



$\therefore \begin{cases} \text{本征值: } \lambda_n = \beta_n^2, n = 1, 2, \dots \\ \text{本征函数: } X_n(x) = B_n \sin \beta_n l, n = 1, 2, 3, \dots \end{cases}$

第3步：求被分离的另一函数 $T(t)$;

将本征值代入 $T'(t) + \lambda a^2 T(t) = 0$ 有

$$T'_n(t) + \beta_n^2 a^2 T_n(t) = 0, \quad n = 1, 2, \dots$$

$$T_n(t) = C'_n e^{-\beta_n^2 a^2 t}, \quad (n = 1, 2, 3, \dots)$$

第4步：叠加特解求一般解；

$$\begin{aligned} u_n(x, t) &= T_n(t) X_n(x) \\ &= C_n e^{-\beta_n^2 a^2 t} \sin \beta_n x, \quad (n = 1, 2, 3, \dots) \end{aligned}$$

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} C_n e^{-\beta_n^2 a^2 t} \sin \beta_n x$$

第5步：根据本征函数的正交性求系数

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-\beta_n^2 a^2 t} \sin \beta_n x$$

由初始条件：

$$u(x, 0) = \sum_{n=1}^{\infty} C_n \sin \beta_n x = \varphi(x), \quad 0 \leq x \leq l$$

$$\int_0^l \varphi(x) \sin \beta_n x dx = \int_0^l \sum_{m=1}^{\infty} C_m \sin \beta_m x \sin \beta_n x dx$$

$$= C_n \int_0^l \sin^2 \beta_n x dx \mapsto C_n L_n$$

$$\therefore C_n = \frac{1}{L_n} \int_0^l \varphi(x) \sin \beta_n x dx$$

$$\therefore \text{解为 } u(x, t) = \sum_{n=1}^{\infty} \left[\frac{1}{L_n} \int_0^l \varphi(x) \sin \beta_n x dx \right] e^{-\beta_n^2 a^2 t} \sin \beta_n x$$

$$X''(x) + \lambda X(x) = 0$$

$$\lambda > 0$$

$$\lambda = 0$$

本征值问题

有限长杆、弦波动和热传导方程

$$X(x) = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x$$

$$X(x) = C$$

$$X(0) = 0$$

$$X'(0) = 0$$

$$C_1 = 0$$

$$C_2 = 0$$

$$X(x) = C_2 \sin(\sqrt{\lambda} x)$$

$$X(x) = C_1 \cos(\sqrt{\lambda} x)$$

$$X(l) = 0 \quad X'(l) = 0$$

$$X(l) = 0 \quad X'(l) = 0$$

$$\sin \sqrt{\lambda} l = 0 \quad \cos \sqrt{\lambda} l = 0 \quad \cos \sqrt{\lambda} l = 0 \quad \sin \sqrt{\lambda} l = 0$$

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2 \quad n=1,2,\dots$$

$$\lambda_n = \left[\frac{(n + \frac{1}{2})\pi}{l} \right]^2 \quad n=0,1,2,\dots$$

泛定方程	边界条件	本征值	本征函数
$u_{tt} = a^2 u_{xx}$ $u_t = a^2 u_{xx}$ $(0 < x < l)$ $(t > 0)$ 本征值问题 常微分方程 $X''(x) + \lambda X(x) = 0$	I,I $u(0,t) = 0$ $u(l,t) = 0$	$\lambda_n = \frac{n^2 \pi^2}{l^2}$ $n = 1, 2, \dots$	$\sin \frac{n\pi}{l} x$ $n = 1, 2, \dots$
	I,II $u(0,t) = 0$ $u_x(l,t) = 0$	$\lambda_n = \left[\frac{(n + \frac{1}{2})\pi}{l} \right]^2$ $n = 0, 1, 2, \dots$	$\sin \left[\frac{(n + \frac{1}{2})\pi}{l} x \right]$ $n = 0, 1, 2, \dots$
	II,I $u_x(0,t) = 0$ $u(l,t) = 0$	$\lambda_n = \left[\frac{(n + \frac{1}{2})\pi}{l} \right]^2$ $n = 0, 1, 2, \dots$	$\cos \left[\frac{(n + \frac{1}{2})\pi}{l} x \right]$ $n = 0, 1, 2, \dots$
	II,II $u_x(0,t) = 0$ $u_x(l,t) = 0$	$\lambda_n = \frac{n^2 \pi^2}{l^2}$ $n = 0, 1, 2, \dots$	$\cos \frac{n\pi}{l} x$ $n = 0, 1, 2, \dots$

$$u_{tt} = a^2 u_{xx}, \quad u_t = a^2 u_{xx}$$

$$\lambda_n = \left[\frac{(n + \frac{1}{2})\pi}{l} \right]^2 \quad \text{I, II}$$

$$X_n(x) = \sin \left[\frac{(n + \frac{1}{2})\pi}{l} x \right]$$

$$n = 0, 1, 2, \dots$$

$$\lambda_n = \frac{n^2 \pi^2}{l^2} \quad \text{I, I}$$

$$X_n(x) = \sin \frac{n\pi}{l} x$$

$$n = 1, 2, \dots$$

$$\lambda_n = \left[\frac{(n + \frac{1}{2})\pi}{l} \right]^2 \quad \text{II, I}$$

$$X_n(x) = \cos \left[\frac{(n + \frac{1}{2})\pi}{l} x \right]$$

$$n = 0, 1, 2, \dots$$

$$\lambda_n = \frac{n^2 \pi^2}{l^2} \quad \text{II, II}$$

$$X_n(x) = \cos \frac{n\pi}{l} x$$

$$n = 0, 1, 2, \dots$$



预备知识

欧拉方程

欧拉方程

形如

$$x^n y^{(n)} + p_1 x^{n-1} y^{(n-1)} + \cdots + p_{n-1} x y' + p_n y = f(x)$$

的方程(其中 $p_1, p_2 \cdots p_n$ 为常数) 叫**欧拉方程**.

特点: 各项未知函数导数的阶数与乘积因子自变量的方次数相同.

解法: 欧拉方程是特殊的变系数方程, 通过变量代换可化为常系数微分方程.

作变量变换 $x = e^t$,

令算子 $D = \frac{d}{dt}$

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} \longrightarrow xy' = Dy$$

$$\frac{d^2 y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \longrightarrow x^2 y'' = D(D-1)y$$

$$\frac{d^3 y}{dx^3} = \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right) \longrightarrow x^3 y''' = D(D-1)(D-2)y$$

.....

$$\dots\dots \longrightarrow x^k y^{(k)} = D(D-1)\cdots(D-k+1)y$$

求以 t 为自变量的常系数线性微分方程，再
把 t 换为 $\ln x$ ，即得到原方程的解。

例 求欧拉方程 $\rho^2 R'' + \rho R' - m^2 R = 0 \quad (m \geq 0)$

解 作变量变换 $\rho = e^t$, 或 $t = \ln \rho$, 原方程化为

$$D(D-1)R + DR - m^2 R = 0$$

$$\rightarrow D^2 R - m^2 R = 0 \rightarrow \frac{d^2 R}{dt^2} - m^2 R = 0$$

特征方程 $r^2 = m^2$, 其根 $r_1 = m, r_2 = -m$

$$m = 0 \text{ 时, } R_0 = C_0 + D_0 t$$

$$m > 0 \text{ 时, } R_m = C_m e^{mt} + D_m e^{-mt}$$

换回原变量 $t = \ln \rho$, 得原方程通解为

$$m = 0 \text{ 时, } R_0 = C_0 + D_0 \ln \rho$$

$$m > 0 \text{ 时, } R_m = C_m \rho^m + D_m \rho^{-m}$$

2.3 二维拉普拉斯方程 的定解问题



一、直角坐标系下的拉普拉斯问题

在矩形域 $0 \leq x \leq a, 0 \leq y \leq b$ 内求Laplace方程

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

边界条件 $\begin{cases} u|_{x=0} = 0, & u|_{x=a} = Ay ; \\ u_y|_{y=0} = 0, & u_y|_{y=b} = 0 . \end{cases}$ II/II类齐次边界条件

解：第1步：分离变量，取试探解 $u(x, y) = X(x)Y(y)$;

$$-\frac{X''(x)}{X(x)} = \frac{Y''(y)}{Y(y)} = -\lambda \Rightarrow \begin{cases} X''(x) - \lambda X(x) = 0 \\ Y''(y) + \lambda Y(y) = 0 \end{cases}$$

第2步：求本征值 λ 和本征函数 $Y(y)$;

由 y 的边界条件得

$$\begin{cases} X(x)Y'(0)=0 \\ X(x)Y'(b)=0 \end{cases} \Rightarrow Y'(0)=Y'(b)=0$$

转化为本征值问题

$$Y''(y) + \lambda Y(y) = 0,$$

边界条件： $Y'(0)=Y'(b)=0$ II/II类齐次边界条件

$$\text{本征值 } \lambda_n = \frac{n^2 \pi^2}{b^2} \quad (n = 0, 1, 2, \dots)$$

$$\text{本征函数 } Y_n(y) = A_n \cos \frac{n\pi}{b} y \quad (n = 0, 1, 2, \dots)$$

第3步：求被分离的另一函数 $X(x)$;

$$X''(x) - \lambda X(x) = 0$$

$$\lambda = 0 \text{ 时, } X_0 = C_0 + D_0 x$$

$$\lambda_n = \frac{n^2 \pi^2}{b^2} \text{ 时, } X_n(x) = C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}} \quad (n = 1, 2, \dots)$$

第4步：叠加特解求一般解;

$$X_0(x) = C_0 + D_0 x, \quad Y_0(y) = A_0,$$

$$X_n(x) = C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}}, \quad Y_n(y) = A_n \cos \frac{n\pi}{b} y,$$

$$u(x, y) = C_0 + D_0 x + \sum_{n=1}^{\infty} \left(C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}} \right) \cos \frac{n\pi}{b} y$$

$$u(x, y) = C_0 + D_0 x + \sum_{n=1}^{\infty} \left(C_n e^{\frac{n\pi x}{b}} + D_n e^{\frac{-n\pi x}{b}} \right) \cos \frac{n\pi}{b} y$$

第5步：由x的边界条件求系数

$$u|_{x=0} = 0 \Rightarrow C_0 + \sum_{n=1}^{\infty} (C_n + D_n) \cos \frac{n\pi y}{b} = 0$$

$$\therefore C_0 = 0, C_n + D_n = 0 \quad (n = 1, 2, \dots)$$

$$u|_{x=a} = Ay \Rightarrow D_0 a + \sum_{n=1}^{\infty} \left(C_n e^{\frac{n\pi a}{b}} + D_n e^{\frac{-n\pi a}{b}} \right) \cos \frac{n\pi}{b} y = Ay$$

根据Ay的Fourier余弦级数展开有

$$D_0 a = \frac{1}{b} \int_0^b Ay dy = \frac{A}{b} \frac{1}{2} b^2 = \frac{Ab}{2} \Rightarrow D_0 = \frac{Ab}{2a}$$

——2.3 二维拉普拉斯方程的定解问题——

$$\begin{aligned} C_n e^{\frac{n\pi a}{b}} + \overset{-C_n}{\uparrow} D_n e^{\frac{-n\pi a}{b}} &= \frac{2}{b} \int_0^b A y \cos \frac{n\pi}{b} y \, dy = \frac{2Ab}{n^2 \pi^2} (\cos n\pi - 1) \\ &= 2C_n \frac{1}{2} (e^{\frac{n\pi a}{b}} - e^{\frac{-n\pi a}{b}}) = 2C_n \operatorname{sh} \left(\frac{n\pi a}{b} \right) \end{aligned}$$

$$\therefore n=0 \text{ 时, } C_0 = 0, D_0 = \frac{Ab}{2a}$$

$$n=1, 2, \dots \text{ 时, } C_n = -D_n = \frac{Ab(\cos n\pi - 1)}{n^2 \pi^2 \operatorname{sh} \frac{n\pi a}{b}}$$

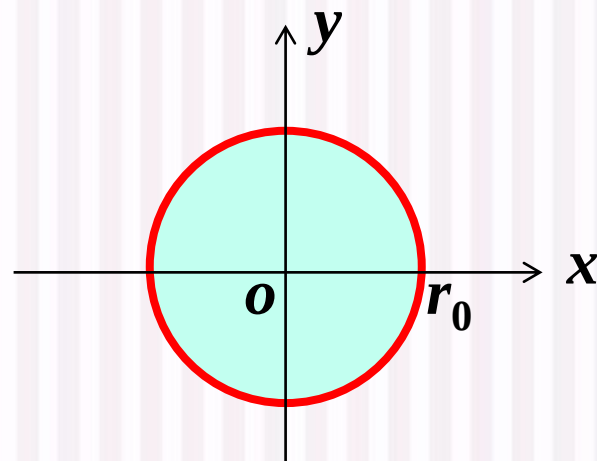
$$\therefore \text{解为 } u(x, y) = \frac{Ab}{2a} x + \frac{2Ab}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos n\pi - 1}{n^2 \operatorname{sh} \frac{n\pi a}{b}} \operatorname{sh} \frac{n\pi x}{b} \cos \frac{n\pi y}{b}$$

二、极坐标系下的拉普拉斯问题

考察一个半径为 r_0 的圆形薄板稳恒状态下的温度分布问题, 设板的上下两面绝热, 圆周边界上的温度已知为 $f(\theta)$, ($0 \leq \theta < 2\pi$), $f(0) = f(2\pi)$.

求稳恒状态下的温度分布规律。

$$\begin{cases} u_{xx} + u_{yy} = 0 & (x^2 + y^2 < r_0^2) \\ u|_{x^2+y^2=r_0^2} = f(\theta) & (0 \leq \theta < 2\pi) \end{cases}$$



作极坐标变换

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}, \begin{cases} r = \sqrt{x^2 + y^2} \\ \varphi = \arctan \frac{y}{x} \end{cases} \Rightarrow \begin{cases} \varphi_x = -\frac{\sin \varphi}{r} \\ \varphi_y = \frac{\cos \varphi}{r} \end{cases}, \begin{cases} r_x = \cos \varphi \\ r_y = \sin \varphi \end{cases}$$

$$\begin{cases} u_{xx} = (u_{rr} r_x + u_{r\varphi} \varphi_x) r_x + u_r r_{xx} + (u_{\varphi\varphi} \varphi_x + u_{\varphi r} r_x) \varphi_x + u_{\varphi} \varphi_{xx} \\ u_{yy} = (u_{rr} r_y + u_{r\varphi} \varphi_y) r_y + u_r r_{yy} + (u_{\varphi\varphi} \varphi_y + u_{\varphi r} r_y) \varphi_y + u_{\varphi} \varphi_{yy} \end{cases}$$

则定解问题可转换为

$$\begin{cases} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2} = 0 & (0 < r < r_0, 0 < \varphi < 2\pi) \\ u|_{r=r_0} = f(\varphi) & (0 \leq \varphi < 2\pi) \end{cases}$$

——2.3 二维拉普拉斯方程的定解问题——

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2} = 0$$

令 $u(r, \varphi) = R(r)\Phi(\varphi)$ 代入方程，得

$$R''(r)\Phi(\varphi) + \frac{R'(r)\Phi(\varphi)}{r} + \frac{R(r)\Phi''(\varphi)}{r^2} = 0$$

两边除以 $R(r)\Phi(\varphi)$ ，整理得

$$\frac{r^2 R''}{R} + \frac{R' r}{R} = \frac{-\Phi''}{\Phi} = \lambda$$

$$\therefore \begin{cases} \Phi'' + \lambda \Phi = 0 \\ r^2 R'' + r R' - \lambda r = 0 \end{cases}$$

——2.3 二维拉普拉斯方程的定解问题——

- 自然周期条件

$$u(r, 0) = u(r, 2\pi) \Rightarrow \Phi(0) = \Phi(2\pi)$$

$$\left. \frac{\partial u(r, \varphi)}{\partial \varphi} \right|_{\varphi=0} = \left. \frac{\partial u(r, \varphi)}{\partial \varphi} \right|_{\varphi=2\pi} \Rightarrow \left. \frac{d\Phi}{d\varphi} \right|_{\varphi=0} = \left. \frac{d\Phi}{d\varphi} \right|_{\varphi=2\pi}$$

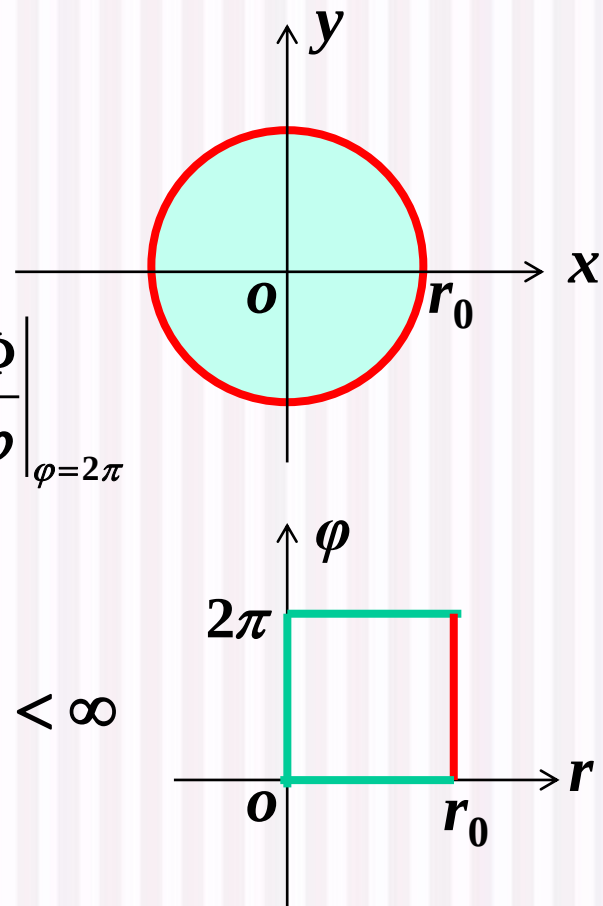
- 有界性条件

$$|u(0, \varphi)| < \infty \Rightarrow |R(0)\Phi(\varphi)| < \infty \Rightarrow |R(0)| < \infty$$

- ◆ 两个常微分方程

$$\begin{cases} \Phi'' + \lambda \Phi = 0 \\ \Phi(0) = \Phi(2\pi) \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$$

$$\begin{cases} r^2 R'' + rR' - \lambda R = 0 \\ |R(0)| < \infty \end{cases}$$



$$\text{圆域本征值问题} \quad \begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(2\pi) = \Phi(0), \quad \left. \frac{d\Phi}{d\varphi} \right|_{\varphi=0} = \left. \frac{d\Phi}{d\varphi} \right|_{\varphi=2\pi} \end{cases}$$

(1) $\lambda < 0$ 时, 通解为 $\Phi(\varphi) = Ae^{\sqrt{-\lambda}\varphi} + Be^{-\sqrt{-\lambda}\varphi}$,

代入边界条件, 则 $A = B = 0$, 无意义.

(2) $\lambda = 0$ 时, 通解为 $\Phi(\varphi) = A\varphi + B$, 由边界条件有

$A = 0$, 得特解 $\Phi(\varphi) = B$.

(3) $\lambda > 0$ 时, 通解 $\Phi(\varphi) = A \cos \sqrt{\lambda}\varphi + B \sin \sqrt{\lambda}\varphi$,

代入边界条件

$$\begin{cases} A \cos 0 + B \sin 0 = A \cos \sqrt{\lambda}(2\pi) + B \sin \sqrt{\lambda}(2\pi) \\ -A\sqrt{\lambda} \sin 0 + B\sqrt{\lambda} \cos 0 = -A\sqrt{\lambda} \sin \sqrt{\lambda}(2\pi) + B\sqrt{\lambda} \cos \sqrt{\lambda}(2\pi) \end{cases}$$

——2.3 二维拉普拉斯方程的定解问题——

$$\begin{cases} A = A \cos 2\pi\sqrt{\lambda} + B \sin 2\pi\sqrt{\lambda} \\ B = -A \sin 2\pi\sqrt{\lambda} + B \cos 2\pi\sqrt{\lambda} \end{cases}$$

若 A, B 不同时为零, 则

$$\begin{vmatrix} \cos(2\pi\sqrt{\lambda}) - 1 & \sin(2\pi\sqrt{\lambda}) \\ -\sin(2\pi\sqrt{\lambda}) & \cos(2\pi\sqrt{\lambda}) - 1 \end{vmatrix} = 0$$

$$\Rightarrow \cos(2\pi\sqrt{\lambda}) = 1 \Rightarrow \lambda = m^2, \quad m = 1, 2, \dots$$

$$\begin{cases} \text{本征值 } \lambda_m = m^2, \quad m = 0, 1, 2, \dots \\ \text{本征函数 } \Phi_m(\varphi) = A_m \cos m\varphi + B_m \sin m\varphi, \quad m = 0, 1, 2, \dots \end{cases}$$

——2.3 二维拉普拉斯方程的定解问题——

$$\begin{cases} r^2 R'' + rR' - \lambda R = 0 \\ |R(0)| < \infty \end{cases}$$

$$r^2 R'' + rR' - m^2 R = 0 \quad \text{----- Euler方程}$$

作变换 $r = e^t$ 得通解为:

$$\left. \begin{array}{l} m = 0 \text{ 时, } R_0 = C_0 + D_0 \ln r \\ m > 0 \text{ 时, } R_m = C_m r^m + D_m r^{-m} \\ |R(0)| < \infty \Rightarrow D_m = 0, \quad m = 0, 1, 2, \dots \end{array} \right\} \begin{array}{l} R_m = C_m r^m \\ m = 0, 1, 2, \dots \end{array}$$

$$\text{特解为: } u_m(r, \varphi) = (A_m \cos m\varphi + B_m \sin m\varphi)(C_m r^m) \\ m = 0, 1, 2, \dots$$

$$\text{叠加得一般解: } u(r, \varphi) = \frac{a_0}{2} + \sum_{m=1}^{\infty} r^m (a_m \cos m\varphi + b_m \sin m\varphi)$$

——2.3 二维拉普拉斯方程的定解问题——

边界条件 $u|_{r=r_0} = f(\varphi) \quad (0 \leq \varphi < 2\pi)$

$$f(\varphi) = u(r_0, \varphi) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (r_0^m a_m \cos m\varphi + r_0^m b_m \sin m\varphi)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(t) dt, \quad a_m = \frac{1}{r_0^m \pi} \int_0^{2\pi} f(t) \cos mt dt$$

$$b_m = \frac{1}{r_0^m \pi} \int_0^{2\pi} f(t) \sin mt dt$$

$$u(r, \varphi) = \frac{a_0}{2} + \sum_{m=1}^{\infty} r^m (a_m \cos m\varphi + b_m \sin m\varphi)$$

$$= \frac{1}{\pi} \int_0^{2\pi} f(t) \left[\frac{1}{2} + \sum_{m=1}^{\infty} \left(\frac{r}{r_0} \right)^m \cos m(\varphi - t) \right] dt$$

$$0 \leq \varphi < 2\pi, \quad r < r_0$$

利用恒等式

$$\frac{1}{2} + \sum_{m=1}^{\infty} k^m \cos m(\varphi - t) = \frac{1}{2} \frac{1 - k^2}{k^2 + 1 - 2k \cos(\varphi - t)}, \quad |k| < 1$$

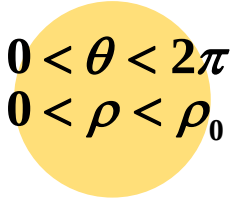
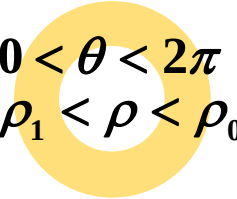
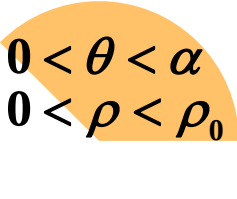
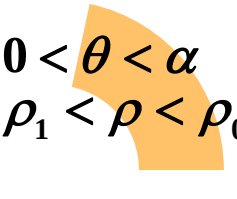
则方程的解可以表示为

$$u(r, \varphi) = \frac{1}{2\pi} \int_0^{2\pi} f(t) \frac{r_0^2 - r^2}{r_0^2 + r^2 - 2r_0 r \cos(\varphi - t)} dt$$
$$(0 \leq \varphi < 2\pi, \quad r < r_0)$$

称为圆域内的泊松公式.

圆环域、扇形域、环扇域等区域上的拉普拉斯方程的边值问题也可用类似方法求解.

二维圆域拉普拉斯问题

区域	边界条件	本征值问题	本征值	本征函数
 $0 < \theta < 2\pi$ $0 < \rho < \rho_0$	$u(\rho_0, \theta) = f(\theta)$	$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(0) = \Phi(2\pi) \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_n = n^2$ $n = 0, 1, 2, \dots$	$A_n \cos n\theta + B_n \sin n\theta$ $n = 0, 1, 2,$
 $0 < \theta < 2\pi$ $\rho_1 < \rho < \rho_0$	$u(\rho_0, \theta) = f_0(\theta)$ $u(\rho_1, \theta) = f_1(\theta)$	$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(0) = \Phi(2\pi) \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_n = n^2$ $n = 0, 1, 2, \dots$	$A_n \cos n\theta + B_n \sin n\theta$ $n = 0, 1, 2,$
 $0 < \theta < \alpha$ $0 < \rho < \rho_0$	$u(\rho_0, \theta) = f(\theta)$ $u(\rho, 0) = 0$ $u(\rho, \alpha) = 0$	$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(0) = \Phi(\alpha) = 0 \end{cases}$	$\lambda = \frac{n^2 \pi^2}{\alpha^2}$ $n = 1, 2, \dots$	$C_n \sin \frac{n\pi}{\alpha} \theta$ $n = 1, 2, \dots$
 $0 < \theta < \alpha$ $\rho_1 < \rho < \rho_0$	$u(\rho_0, \theta) = f_0(\theta)$ $u(\rho_1, \theta) = f_1(\theta)$ $u(\rho, 0) = u(\rho, \alpha) = 0$	$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(0) = \Phi(\alpha) = 0 \end{cases}$	$\lambda = \frac{n^2 \pi^2}{\alpha^2}$ $n = 1, 2, \dots$	$C_n \sin \frac{n\pi}{\alpha} \theta$ $n = 1, 2, \dots$



2.4 非齐次方程的解法

一、弦的强迫振动

二、极坐标系求解非齐次方程

一、弦的强迫振动 ----- 非齐次方程，齐次边界条件

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + f(x, t), & 0 < x < l, t > 0 \\ u(0, t) = u(l, t) = 0, & t > 0 \\ u(x, 0) = \varphi(x), \quad \frac{\partial u(x, 0)}{\partial t} = \psi(x), & 0 \leq x \leq l \end{cases}$$

由线性方程的叠加原理, 令: $u(x, t) = v(x, t) + w(x, t)$

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2} \\ v(0, t) = v(l, t) = 0, \\ v(x, 0) = \varphi(x), \quad \frac{\partial v(x, 0)}{\partial t} = \psi(x) \end{cases}$$

初始状态引起的弦的位移

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} = a^2 \frac{\partial^2 w}{\partial x^2} + f(x, t), & 0 < x < l, t > 0, \\ w(0, t) = w(l, t) = 0, & t > 0, \\ w(x, 0) = \frac{\partial w(x, 0)}{\partial t} = 0, & 0 \leq x \leq l, \end{cases}$$

强迫力引起的弦的位移

仅由初始状态引起的弦的位移

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2} & \longrightarrow \text{齐次方程} \\ v(0, t) = v(l, t) = 0, & \longrightarrow \text{齐次边界条件} \\ v(x, 0) = \varphi(x), \frac{\partial v(x, 0)}{\partial t} = \psi(x) & \longrightarrow \text{非零初始条件} \end{cases}$$

其解为一系列驻波叠加而成：

$$v(x, t) = \sum_{n=1}^{\infty} \left[\left(\frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi}{l} x dx \right) \cos \frac{n\pi a}{l} t + \left(\frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi}{l} x dx \right) \sin \frac{n\pi a}{l} t \right] \sin \frac{n\pi}{l} x$$

仅由强迫力引起的弦的位移

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} = a^2 \frac{\partial^2 w}{\partial x^2} + f(x, t), & 0 < x < l, t > 0, & \text{非齐次方程} \\ w(0, t) = w(l, t) = 0, & t > 0, & \text{齐次边界条件} \\ w(x, 0) = \frac{\partial w(x, 0)}{\partial t} = 0, & 0 \leq x \leq l, & \text{零初始条件} \end{cases}$$

假定其解也可按前述本征函数展开:

$$w(x, t) = \sum_{n=1}^{\infty} w_n(t) \sin \frac{n\pi}{l} x, \quad w_n(t) \text{ 为待定函数.}$$

$$\text{初始条件 } w(x, 0) = \sum_{n=1}^{\infty} w_n(0) \sin \frac{n\pi}{l} x = 0 \Rightarrow w_n(0) = 0$$

$$\frac{\partial w(x, 0)}{\partial t} = \sum_{n=1}^{\infty} w'_n(0) \sin \frac{n\pi}{l} x = 0 \Rightarrow w'_n(0) = 0$$

将 $f(x, t)$ 也按照本征函数系展开:

$$f(x, t) = \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi}{l} x, \text{ 其中 } f_n(t) = \frac{2}{l} \int_0^l f(x, t) \sin \frac{n\pi}{l} x dx$$

代入非齐次波动方程有

$$\sum_{n=1}^{\infty} w_n''(t) \sin \frac{n\pi}{l} x = \sum_{n=1}^{\infty} \left[-a^2 \frac{n^2 \pi^2}{l^2} w_n(t) \sin \frac{n\pi}{l} x \right] + \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi}{l} x$$

二阶线性非齐次常微分方程问题

$$w_n''(t) + a^2 \frac{n^2 \pi^2}{l^2} w_n(t) = f_n(t)$$

$$\text{初始条件 } w_n(0) = 0, w_n'(0) = 0$$

$$w_n''(t) + a^2 \frac{n^2 \pi^2}{l^2} w_n(t) = f_n(t)$$

$$\text{初始条件 } w_n(0) = 0, w_n'(0) = 0$$

用拉普拉斯变换法求此常微分方程

$$\text{令 } L[w_n(t)] = W_n(s), L[f_n(t)] = F_n(s)$$

$$L[w_n'(t)] = sW_n(s) - w_n(0) = sW_n(s)$$

$$L[w_n''(t)] = s(sW_n(s) - w_n(0)) - w_n'(0) = s^2 W_n(s)$$

$$\text{则方程变换为: } s^2 W_n(s) + a^2 \frac{n^2 \pi^2}{l^2} W_n(s) = F_n(s)$$

$$\therefore W_n(s) = \frac{1}{s^2 + a^2 \frac{n^2 \pi^2}{l^2}} F_n(s)$$

作拉氏反变换

$$L^{-1}\left[\frac{k}{s^2 + k^2}\right] = \sin kt \Rightarrow L^{-1}\left[\frac{1}{s^2 + a^2 \frac{n^2 \pi^2}{l^2}}\right] = \frac{l}{an\pi} \sin \frac{an\pi}{l} t$$

$$\therefore w_n(t) = \frac{l}{n\pi a} \sin \frac{n\pi a}{l} t * f_n(t) = \frac{l}{n\pi a} \int_0^t f_n(\tau) \sin \frac{n\pi a}{l} (t - \tau) d\tau$$

$$\therefore w(x, t) = \sum_{n=1}^{\infty} \left[\frac{l}{n\pi a} \int_0^t f_n(\tau) \sin \frac{n\pi a}{l} (t - \tau) d\tau \right] \sin \frac{n\pi}{l} x$$

$\therefore$ 原非齐次PDE方程的解为

$$u(x, t) = w(x, t) + v(x, t)$$

本征函数法

非齐次方程 u (齐次边界条件)

齐次方程 v
(齐次边界条件, 初始条件)

+

非齐次方程 w
(齐次边界条件, 零初始条件)



直接用分离变量法

先求本征值和本征函数,
再确定系数



本征函数展开法

把非齐次方程的解和非齐次项都
按齐次方程相应的本征函数展开



叠加 $u = v + w$

直接本征函数展开法

若本征函数已知, 可直接将非齐次项和初始条件也按本征函数系展开, 然后求解常微分方程的非齐次初始值问题。

例 求解具有放射衰变的热传导方程

$$\frac{\partial u}{\partial t} = b^2 \frac{\partial^2 u}{\partial x^2} + B e^{-\beta x}$$

初始条件 $u|_{t=0} = T_0$ (常数)

边界条件 $u|_{x=0} = 0, \quad u|_{x=l} = 0$ I/I类齐次边界条件

$$u_{tt} = a^2 u_{xx}$$

$$u_t = a^2 u_{xx}$$

$$u(0, t) = 0, \quad u(l, t) = 0$$



$$\text{本征值: } \lambda_n = \frac{n^2 \pi^2}{l^2}$$

$$\text{本征函数: } X_n(x) = \sin \frac{n\pi}{l} x$$

$$n = 1, 2, \dots$$

直接按本征函数展开：

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi}{l} x$$

$$Be^{-\beta x} = \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi}{l} x$$

$$T_0 = \sum_{n=1}^{\infty} T_n(0) \sin \frac{n\pi}{l} x$$

$$\begin{aligned} f_n(t) &= \frac{2}{l} \int_0^l Be^{-\beta \xi} \sin \frac{n\pi}{l} \xi d\xi \\ &= \frac{2n\pi B [1 - (-1)^n e^{-\beta l}]}{\beta^2 + (n\pi)^2} \end{aligned}$$

$$\begin{aligned} T_n(0) &= \frac{2}{l} \int_0^l T_0 \sin \frac{n\pi}{l} \xi d\xi \\ &= \frac{2T_0}{n\pi} [1 - (-1)^n] \end{aligned}$$

代入到原方程和初始条件得

$$\left\{ \begin{aligned} &\sum_{n=1}^{\infty} \left[T_n'(t) + \left(\frac{n\pi b}{l} \right)^2 T_n(t) \right] \sin \frac{n\pi}{l} x = \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi}{l} x \\ &\text{初始条件: } \sum_{n=1}^{\infty} T_n(0) \sin \frac{n\pi}{l} x = T_0 \end{aligned} \right.$$

$$\begin{cases} T_n'(t) + \left(\frac{n\pi b}{l} \right)^2 T_n(t) = \frac{2n\pi B [1 - (-1)^n e^{-\beta l}]}{\beta^2 + (n\pi)^2} \\ \text{初始条件: } T_n(0) = \frac{2T_0}{n\pi} [1 - (-1)^n] \end{cases}$$

知识回顾：一阶线性非齐次常微分方程

$y' + P(x)y = Q(x)$ 的解：

$$y = e^{-\int_{x_0}^x P(x)dx} \left[y_0 + \int_{x_0}^x Q(x) e^{\int_{x_0}^x P(x)dx} dx \right]$$

利用一阶常微分方程的解：

$$T_n(t) = \frac{2T_0}{n\pi} [1 - (-1)^n] e^{-\left(\frac{n\pi b}{l}\right)^2 t} + \frac{2B}{b^2 n\pi} \frac{[1 - (-1)^n e^{-\beta l}]}{\beta^2 + \left(\frac{n\pi}{l}\right)^2} \left[1 - e^{-\left(\frac{n\pi b}{l}\right)^2 t} \right]$$

于是原定解问题的解为

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi}{l} x, \quad T_n(t) \text{ 如上}.$$

二、极坐标系求解非齐次方程

例 在环形域 $a \leq \sqrt{x^2 + y^2} \leq b$ ($0 < a < b$) 内求解
下列定解问题.

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 12(x^2 - y^2) & (a < \sqrt{x^2 + y^2} < b), \\ u|_{\sqrt{x^2 + y^2} = a} = 0, \quad \frac{\partial u}{\partial \vec{n}} \Big|_{\sqrt{x^2 + y^2} = b} = 0. & \text{I/II类齐次边界条件} \end{cases}$$

(求解区域是环形区域, 宜选用平面极坐标系.)

转换为极坐标系

$$\begin{cases} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 12\rho^2 \cos 2\varphi, & (a < \rho < b), \\ u|_{\rho=a} = 0, & \left. \frac{\partial u}{\partial \rho} \right|_{\rho=b} = 0. \end{cases}$$

解:采用本征函数法. 由圆域Laplace方程的本征函数 $\Phi_m(\varphi) = A_m \cos m\varphi + B_m \sin m\varphi$, 令解的形式为:

$$\begin{aligned} u(\rho, \varphi) &= \sum_{m=0}^{\infty} R_m(\rho) \Phi_m(\varphi) \\ &= \sum_{m=0}^{\infty} [A_m(\rho) \cos m\varphi + B_m(\rho) \sin m\varphi] \end{aligned}$$

将解代入边界条件得

$$\begin{aligned} u(a, \varphi) &= \sum_{m=0}^{\infty} [A_m(a) \cos m\varphi + B_m(a) \sin m\varphi] = 0 \\ \Rightarrow A_m(a) &= B_m(a) = 0 \end{aligned} \quad (1)$$

$$\begin{aligned} \left. \frac{\partial u}{\partial \rho} \right|_{\rho=b} &= \sum_{m=0}^{\infty} [A'_m(b) \cos m\varphi + B'_m(b) \sin m\varphi] = 0 \\ \Rightarrow A'_m(b) &= B'_m(b) = 0 \end{aligned} \quad (2)$$

将解代入方程得

$$\sum_{m=0}^{\infty} \left\{ \left[A_m''(\rho) + \frac{1}{\rho} A_m'(\rho) - \frac{m^2}{\rho^2} A_m(\rho) \right] \cos m\varphi + \left[B_m''(\rho) + \frac{1}{\rho} B_m'(\rho) - \frac{m^2}{\rho^2} B_m(\rho) \right] \sin m\varphi \right\} = 12\rho^2 \cos 2\varphi.$$

比较两端系数，可得

$$\left. \begin{aligned} A_2''(\rho) + \frac{1}{\rho} A_2'(\rho) - \frac{4}{\rho^2} A_2(\rho) &= 12\rho^2 & (3) \\ A_m''(\rho) + \frac{1}{\rho} A_m'(\rho) - \frac{m^2}{\rho^2} A_m(\rho) &= 0 \quad (m \neq 2) & (4) \\ B_m''(\rho) + \frac{1}{\rho} B_m'(\rho) - \frac{m^2}{\rho^2} B_m(\rho) &= 0 & (5) \end{aligned} \right\} \text{欧拉方程}$$

三个欧拉方程的通解分别为

$$A_2(\rho) = \rho^4 + c_2\rho^2 + d_2\rho^{-2}$$

$$A_m(\rho) = c_m\rho^m + d_m\rho^{-m} \quad (m \neq 2)$$

$$B_m(\rho) = e_m\rho^m + f_m\rho^{-m}$$

$$\text{由 } A_m(a) = B_m(a) = A'_m(b) = B'_m(b) = 0$$

$$\left. \begin{aligned} A_m(a) &= c_m a^m + d_m a^{-m} = 0 \\ A'_m(b) &= m c_m b^{m-1} - m d_m b^{-m-1} = 0 \end{aligned} \right\} \Rightarrow \begin{cases} c_m = 0 \\ d_m = 0 \end{cases}$$

$$\therefore A_m(\rho) \equiv 0 \quad (m \neq 2),$$

$$\text{同理 } B_m(\rho) \equiv 0 \quad (m = 1, 2, \dots).$$

——2.4 非齐次方程的解法——

$$m = 2 \text{ 时, } A_2(\rho) = \rho^4 + c_2 \rho^2 + d_2 \rho^{-2},$$

$$A_2(a) = A_2'(b) = 0$$

$$\left. \begin{aligned} A_2(a) &= a^4 + c_2 a^2 + d_2 a^{-2} = 0 \\ A_2'(b) &= 4b^3 + 2c_2 b - 2d_2 b^{-3} = 0 \end{aligned} \right\} \Rightarrow \begin{cases} c_2 = -\frac{a^6 + 2b^6}{a^4 + b^4} \\ d_2 = -\frac{a^4 b^4 (a^2 - 2b^2)}{a^4 + b^4} \end{cases}$$

$$\therefore A_2(\rho) = \rho^4 - \frac{a^6 + 2b^6}{a^4 + b^4} \rho^2 - \frac{a^4 b^4 (a^2 - 2b^2)}{a^4 + b^4} \rho^{-2}$$

原定解问题的解为

$$\begin{aligned} u(\rho, \varphi) &= \sum_{m=0}^{\infty} [A_m(\rho) \cos m\varphi + B_m(\rho) \sin m\varphi] \\ &= A_2(\rho) \cos 2\varphi \end{aligned}$$



2.5 非齐次边界条件的处理

- 一、非齐次方程(非齐次边界条件)
→ 非齐次方程(齐次边界条件)
- 二、非齐次方程(非齐次边界条件)
→ 齐次方程(齐次边界条件)

非齐次边界条件的处理

基本原则：令 $u(x,t) = V(x,t) + W(x,t)$

适当选取辅助函数 $W(x,t)$,

使得对 $V(x,t)$ 边界条件为齐次的,

某些条件下方程也可化为齐次的.

一、非齐次方程(非齐次边界条件)

→非齐次方程(齐次边界条件)

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + f(x, t), & 0 < x < l, t > 0 \\ u(0, t) = u_1(t), u(l, t) = u_2(t), & t > 0 \\ u(x, 0) = \varphi(x), \frac{\partial u(x, 0)}{\partial t} = \psi(x), & 0 \leq x \leq l \end{cases}$$

解: 令 $u(x, t) = V(x, t) + W(x, t)$, 其中 $V(0, t) = 0, V(l, t) = 0$,
且 $W(0, t) = u_1(t), W(l, t) = u_2(t)$.

取辅助函数 $W(x, t) = A(t)x + B(t)$, 则

$$\left. \begin{array}{l} W(0, t) = B(t) = u_1(t) \\ W(l, t) = A(t)l + B(t) = u_2(t) \end{array} \right\} W(x, t) = \frac{u_2(t) - u_1(t)}{l} x + u_1(t)$$

——2.5 非齐次边界条件的处理——

$$u(x, t) = V(x, t) + W(x, t) = V(x, t) + u_1(t) + \frac{1}{l} [u_2(t) - u_1(t)] x$$
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + f(x, t), \\ u(0, t) = u_1(t), u(l, t) = u_2(t); u(x, 0) = \varphi(x), \frac{\partial u(x, 0)}{\partial t} = \psi(x) \end{cases}$$

可转换为关于V的非齐次方程(齐次边界条件):

$$\begin{cases} \frac{\partial^2 V}{\partial t^2} = a^2 \frac{\partial^2 V}{\partial x^2} + f(x, t) - \frac{1}{l} [u_2''(t) - u_1''(t)] x - u_1''(t) \\ V(0, t) = 0, V(l, t) = 0, \\ V(x, 0) = \varphi(x) - \frac{1}{l} [u_2(0) - u_1(0)] x - u_1(0), \\ \frac{\partial V(x, 0)}{\partial t} = \psi(x) - \frac{1}{l} [u_2'(0) - u_1'(0)] x - u_1'(0) \end{cases}$$

常见非齐次边界条件齐次化所使用辅助函数

非齐次边界条件	齐次化所使用辅助函数
$u(0, t) = u_1(t), u(l, t) = u_2(t)$	$W(x, t) = \frac{1}{l} [u_2(t) - u_1(t)] x + u_1(t)$
$u(0, t) = u_1(t), u_x(l, t) = u_2(t)$	$W(x, t) = u_2(t)x + u_1(t)$
$u_x(0, t) = u_1(t), u(l, t) = u_2(t)$	$W(x, t) = u_1(t)(x - l) + u_2(t)$
$u_x(0, t) = u_1(t), u_x(l, t) = u_2(t)$	$W(x, t) = \frac{u_2(t) - u_1(t)}{2l} x^2 + u_1(t)x$

以上方法适用于波动方程、热传导方程和拉普拉斯方程。

$$\text{例: } \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = \sin \omega t, & t > 0 \\ u(x, 0) = 0, \frac{\partial u(x, 0)}{\partial t} = 0, & 0 \leq x \leq l \end{cases}$$

变换1: 令 $u(x, t) = V(x, t) + W(x, t)$,

$$W(x, t) = \frac{u_2(t) - u_1(t)}{l} x + u_1(t) = \frac{x}{l} \sin \omega t$$

$$\begin{cases} V_{tt} = a^2 V_{xx} + \frac{\omega^2}{l} x \sin \omega t, & \longrightarrow \text{非齐次方程} \\ V(0, t) = 0, V(l, t) = 0, \\ V(x, 0) = 0, V_t(x, 0) = -\frac{\omega}{l} x. \end{cases}$$



容我做个悲伤的表情

变换2: 令 $u(x, t) = V(x, t) + W(x, t)$,

$$\text{取 } W(x, t) = \frac{\sin \frac{\omega}{a} x}{\sin \frac{\omega}{a} l} \sin \omega t,$$

则原问题可转换为:

$$\begin{cases} V_{tt} = a^2 V_{xx}, \\ V(0, t) = 0, \quad V(l, t) = 0, \\ V(x, 0) = 0, \quad V_t(x, 0) = -\frac{\omega \sin \frac{\omega}{a} x}{\sin \frac{\omega}{a} l}. \end{cases} \longrightarrow \text{齐次方程}$$

二、非齐次方程(非齐次边界条件) →齐次方程(齐次边界条件)

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + f(x, t), & 0 < x < l, t > 0 \\ u(0, t) = u_1(t), u(l, t) = u_2(t), & t > 0 \\ u(x, 0) = \varphi(x), \frac{\partial u(x, 0)}{\partial t} = \psi(x), & 0 \leq x \leq l \end{cases}$$

令 $u(x, t) = V(x, t) + W(x, t)$,

$$\begin{cases} \frac{\partial^2 V}{\partial t^2} = a^2 \frac{\partial^2 V}{\partial x^2} + f(x, t) + a^2 \frac{\partial^2 W}{\partial x^2} - \frac{\partial^2 W}{\partial t^2}, \\ V(0, t) = u_1(t) - W(0, t), \quad V(l, t) = u_2(t) - W(l, t), \\ V(x, 0) = \varphi(x) - W(x, 0), \quad \frac{\partial V(x, 0)}{\partial t} = \psi(x) - \frac{\partial W}{\partial t}(x, 0) \end{cases}$$

$$\begin{cases} f(x, t) + a^2 \frac{\partial^2 W}{\partial x^2} - \frac{\partial^2 W}{\partial t^2} = 0, \\ W(0, t) = u_1(t), \\ W(l, t) = u_2(t), \end{cases}$$

若能从中求出 $W(x, t)$,
就可以实现两者
同时齐次化.
但一般很
难求出!



若 $f(x, t)$ 和非齐次边界条件 $u_1(t)$ 和 $u_2(t)$ 都与 t 无关, 则此时 W 仅是 x 的函数 $W(x)$, 即

$$\begin{cases} a^2 W'' + f(x) = 0, \\ W(0) = u_1, \\ W(l) = u_2, \end{cases} \quad \text{二阶常微分方程, 易求!}$$

同时齐次化条件: $f(x,t)$, $u_1(t)$ 和 $u_2(t)$ 都与 t 无关

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + A, & 0 < x < l, \quad t > 0; \\ u|_{x=0} = 0, \quad u|_{x=l} = B; \\ u|_{t=0} = 0, \quad \frac{\partial u}{\partial t}|_{t=0} = 0 \end{cases} \quad \text{其中 } A, B \text{ 为常数.}$$

解: 令 $u(x,t) = v(x,t) + w(x)$, 则 $v_{tt} = a^2[v_{xx} + w''(x)] + A$

$$\begin{cases} a^2 w''(x) + A = 0 \\ w|_{x=0} = 0, \quad w|_{x=l} = B \end{cases}$$

$$\therefore w(x) = -\frac{A}{2a^2} x^2 + \left(\frac{Al}{2a^2} + \frac{B}{l} \right) x$$

——2.5 非齐次边界条件的处理——

$$u(x, t) = v(x, t) - \frac{A}{2a^2} x^2 + \left(\frac{Al}{2a^2} + \frac{B}{l} \right) x$$

则原问题
$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} + A, & 0 < x < l, \quad t > 0; \\ u|_{x=0} = 0, \quad u|_{x=l} = B; \quad u|_{t=0} = 0, \quad \frac{\partial u}{\partial t}|_{t=0} = 0 \end{cases}$$

可转换为关于 v 的齐次方程 (齐次边界条件):

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}, & 0 < x < l, \quad t > 0; \\ v|_{x=0} = 0, \quad v|_{x=l} = 0; \\ v|_{t=0} = -w(x), \quad \frac{\partial v}{\partial t}|_{t=0} = 0 \end{cases}$$

利用分离变量法求解得

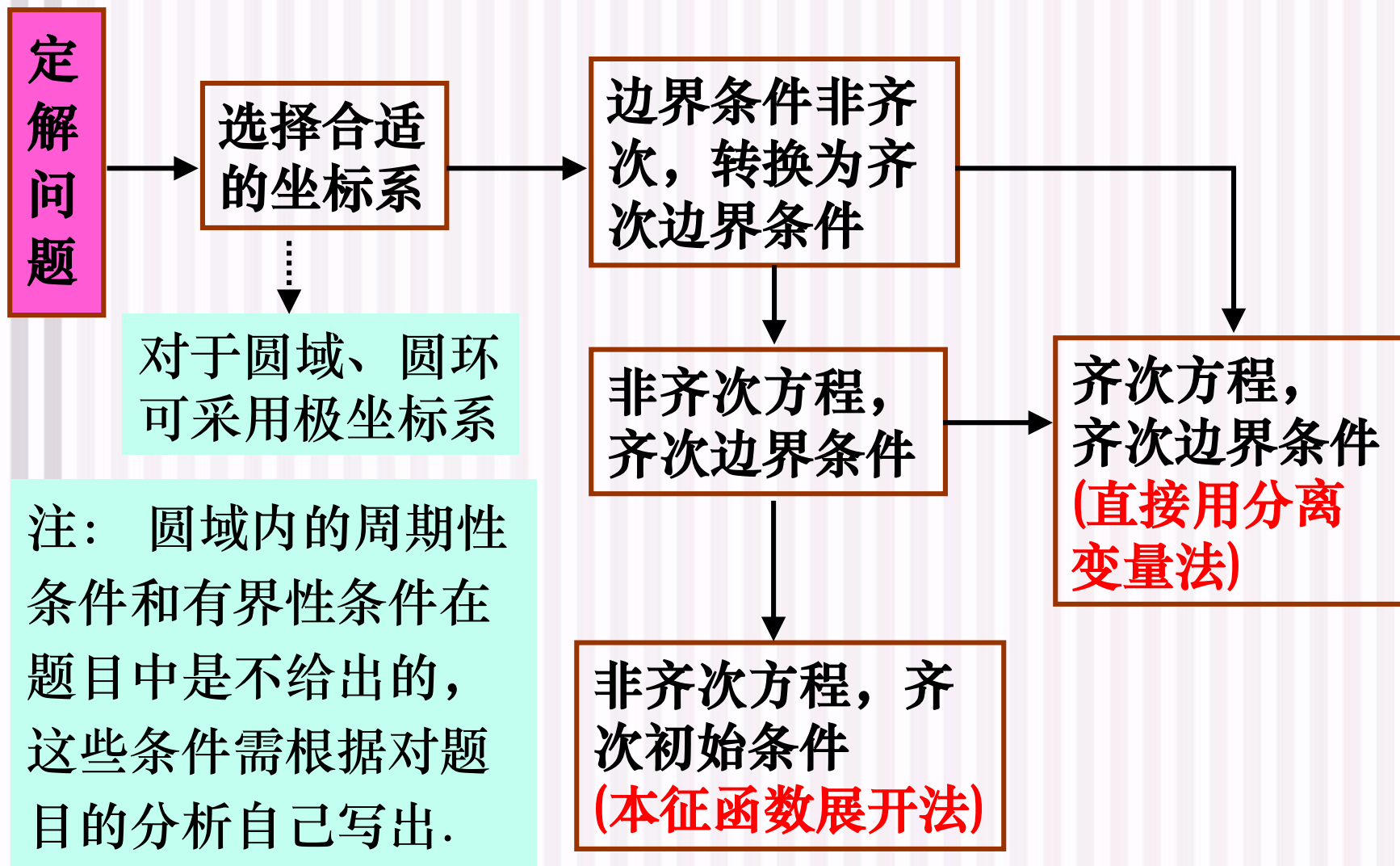
$$v(x, t) = \sum_{n=1}^{\infty} \left(C_n \cos \frac{na\pi}{l} t + D_n \sin \frac{na\pi}{l} t \right) \sin \frac{n\pi}{l} x$$

$$\text{其中 } C_n = -\frac{2Al^2}{a^2 n^3 \pi^3} + \frac{2}{n\pi} \left(\frac{Al^2}{a^2 n^2 \pi^2} + B \right) \cos n\pi, \quad D_n = 0.$$

从而，原定解问题的解为

$$\begin{aligned} u(x, t) &= v(x, t) + w(x) \\ &= -\frac{A}{2a^2} x^2 + \left(\frac{Al}{2a^2} + \frac{B}{l} \right) x + \sum_{n=1}^{\infty} C_n \cos \frac{na\pi}{l} t \sin \frac{n\pi}{l} x \end{aligned}$$

应用分离变量法求解定解问题



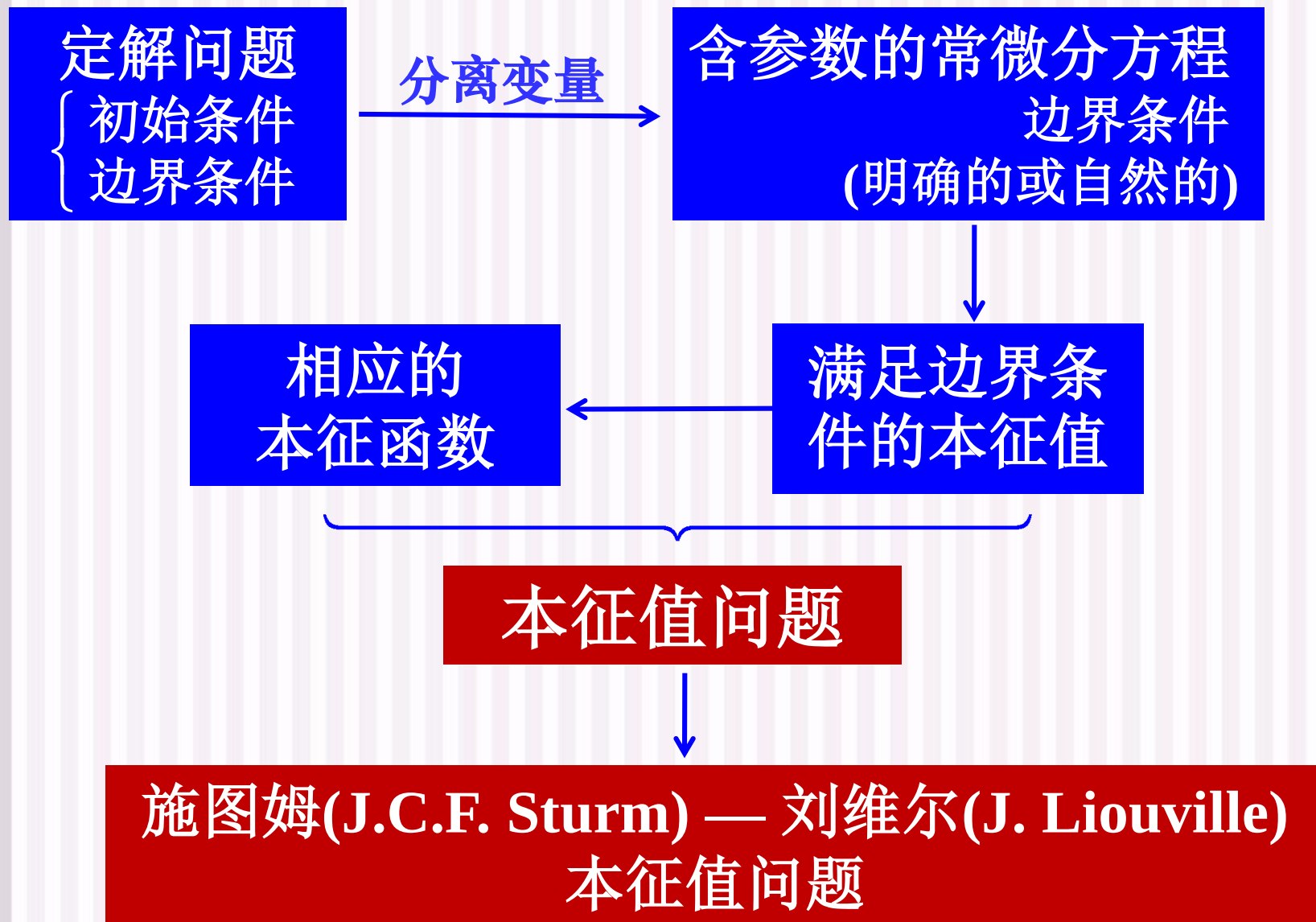


2.6 本征值问题

2.6.1 施图姆-刘维尔方程 (Sturm-Liouville)

2.6.2 本征值问题的一般提法

2.6.3 本征值问题的一般性质





2.6.1 施图姆-刘维尔方程

1. 定义

$$\text{形如 } \frac{d}{dx} \left[k(x) \frac{dy}{dx} \right] - \gamma(x)y + \lambda \rho(x)y = 0$$

其中 $a \leq x \leq b$, $k(x) \geq 0$, $\gamma(x) \geq 0$, $\rho(x) \geq 0$,

λ 为参数, $\rho(x)$ 为权函数, $k(x)$ 为核函数

方程称为施图姆-刘维尔方程, 简称S-L型方程.

$$\text{引入S-L算符: } L \equiv -\frac{d}{dx} \left[k(x) \frac{d}{dx} \right] + \gamma(x)$$

则S-L方程可写为 $L[y] = \lambda \rho(x)y$

2. 二阶线性ODE $\rightarrow$ S-L方程

$$y'' + a(x)y' + b(x)y + \lambda c(x)y = 0$$

S-L标准方程: $\frac{d}{dx} \left[k(x) \frac{dy}{dx} \right] - \gamma(x)y + \lambda \rho(x)y = 0 \quad \Leftrightarrow$

$$y'' + \frac{k'(x)}{k(x)}y' - \frac{\gamma(x)}{k(x)}y + \frac{\lambda \rho(x)}{k(x)}y = 0$$

$$a(x) = \frac{k'(x)}{k(x)} \Leftrightarrow \int a(x)dx = \int \frac{1}{k(x)} dk(x) \Leftrightarrow k(x) = e^{\int a(x)dx}$$

$$k(x) = e^{\int a(x)dx}, \quad \gamma(x) = -b(x)k(x), \quad \rho(x) = c(x)k(x).$$

$$\text{S-L方程 } \frac{d}{dx} \left[k(x) \frac{dy}{dx} \right] - \gamma(x)y + \lambda \rho(x)y = 0$$

◆ 波动方程的本征方程 $X'' + \lambda X = 0$

可写为
$$\frac{d}{dx} \left[\frac{dX}{dx} \right] + \lambda X = 0$$

其中 $k(x) = 1$, $\gamma(x) = 0$, $\rho(x) = 1$, $\lambda = \lambda$ (为参数).

◆ Bessel方程 $x^2 y'' + xy' + (m^2 x^2 - \mu^2)y = 0$

可写为
$$\frac{d}{dx} \left[x \frac{dy}{dx} \right] - \frac{\mu^2}{x} y + m^2 xy = 0$$

其中 $k(x) = e^{\int \frac{1}{x} dx} = x$, $\gamma(x) = -\frac{\mu^2}{x^2} x = \frac{\mu^2}{x}$,
 $\rho(x) = x$, $\lambda = m^2$ (为参数).



2.6.2 本征值问题的一般提法

1 施图姆-刘维尔本征值问题

$$\begin{cases} \frac{d}{dx} \left[k(x) \frac{dy}{dx} \right] - \gamma(x)y + \lambda \rho(x)y = 0, & a \leq x \leq b \\ \text{边界条件 (第一、二、三类或自然边界条件)} \end{cases}$$

在一定的边界条件下，只有当本征值 λ 取某些特定的值时，才有满足边界条件的非零解。

本征值的全体称为给定问题的“谱”。

2 S-L本征值问题边界条件的三种提法

(1) 第一、二、三类边界条件

若 $k(a) \neq 0 (a \in \partial\Omega)$, 则应附加齐次边界条件

$\alpha y'(a) + \beta y(a) = 0$, α 和 β 为常数, 且不同时为零.

例如: 第一类边界条件(此时 $\alpha = 0$)

$$\begin{cases} y''(x) + \lambda y(x) = 0, & (0 \leq x \leq l) \\ y(0) = 0, & y(l) = 0 \end{cases}$$

再如: 第一、三类边界条件

$$\begin{cases} y''(x) + \lambda y(x) = 0, & (0 \leq x \leq l) \\ y(0) = 0, & y'(l) + hy(l) = 0 \end{cases}$$

(2) 有界条件

若 $k(a) = 0$ 且是 $k(x)$ 的一级零点,

则应附加边界条件: $y(a) \neq \infty$, 即有界.

证明: $k(a)$ 是一级零点 $\Rightarrow k(a) = 0, k'(a) \neq 0$

设 $y_1(x)$ 是S-L方程的一个解且 $y_1(a) \neq \infty$,

由Liouville公式有

$$\text{S-L方程: } y'' + \frac{k'(x)}{k(x)} y' + [\lambda \rho(x) - \gamma(x)] y = 0$$

$$y_2(x) = y_1(x) \left[\int_{x_0}^x \frac{1}{y_1^2(\xi)} e^{-\int \frac{k'(\xi)}{k(\xi)} d\xi} d\xi \right] = y_1(x) \left[\int_{x_0}^x \frac{1}{k(\xi) y_1^2(\xi)} d\xi \right]$$

$$k(a) = 0 \Rightarrow y_2(a) = \infty$$

所以应附加边界条件 $y(a) \neq \infty$

例如: Bessel方程本征值问题

$$\begin{cases} \frac{d}{dx} \left[x \frac{dy}{dx} \right] - \frac{\mu^2}{x} y + m^2 xy = 0, & 0 \leq x \leq a \\ y(0) \neq \infty, [\alpha y'(x) + \beta y(x)] \Big|_{x=a} = 0, & \alpha, \beta \text{ 为常数.} \end{cases}$$

又如: Legendre方程的本征值问题

$$\begin{cases} (1-x^2)y''(x) - 2xy'(x) + \lambda y(x) = 0 & (-1 \leq x \leq 1) \\ |y(\pm 1)| < +\infty \end{cases}$$

(3) 周期条件

若 $k(a) = k(b)$ ，则应附加周期性边界条件：

$$y(a) = y(b), \quad y'(a) = y'(b)$$

例如：本征值问题

$$y''(x) + \lambda y(x) = 0, \quad 0 \leq x \leq 2\pi$$

$$\text{边界条件： } y(0) = y(2\pi), \quad y'(0) = y'(2\pi)$$

有界条件和周期条件都可称为自然边界条件.



2.6.3 本征值问题的一般性质

性质1(可数性) 如果 $k(x)$ 及其一阶导数连续, $\gamma(x)$ 或连续或在边界上有一级极点, 则S-L本征值问题有无穷多个分立的**实本征值**.

$$\lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \lambda_n \leq \cdots$$

若 λ_0 不存在, 则从 λ_1 开始, 相应地有本征函数

$$y_1(x), y_2(x), \cdots, y_n(x), \cdots.$$

性质2 (非负性) 所有本征值均为实数且 $\lambda_i \geq 0$.

性质3 (正交性) 对应于不同本征值 λ_m 和 λ_n 的本征函数 $y_m(x)$ 与 $y_n(x)$ 在区间 $[a, b]$ 上加权正交, 即

$$\int_a^b y_m(x)y_n(x)\rho(x)dx = 0 \quad (m \neq n).$$

证明 因为 $\frac{d}{dx}\left[k \frac{dy_m}{dx}\right] - \gamma y_m + \lambda_m \rho y_m = 0$

两函数加权内积, 记为 (y_m, y_n)

$$\frac{d}{dx}\left[k \frac{dy_n}{dx}\right] - \gamma y_n + \lambda_n \rho y_n = 0$$

上式乘以 y_n , 下式乘以 y_m , 两式相减并积分, 得

$$\int_a^b \left[y_n \frac{d}{dx} \left(k \frac{dy_m}{dx} \right) - y_m \frac{d}{dx} \left(k \frac{dy_n}{dx} \right) \right] dx + (\lambda_m - \lambda_n) \int_a^b y_m y_n \rho dx = 0$$

$$\begin{aligned} \text{其中 } & \int_a^b \left[y_n \frac{d}{dx} \left(k \frac{dy_m}{dx} \right) - y_m \frac{d}{dx} \left(k \frac{dy_n}{dx} \right) \right] dx \\ &= y_n k y'_m \Big|_a^b - \int_a^b y'_n k y'_m dx - y_m k y'_n \Big|_a^b + \int_a^b y'_m k y'_n dx \\ &= k(b) [y_n(b) y'_m(b) - y_m(b) y'_n(b)] \\ &\quad - k(a) [y_n(a) y'_m(a) - y_m(a) y'_n(a)] \end{aligned}$$

可以证明上式在三种齐次边界条件下都为零，因此

当 $\lambda_m \neq \lambda_n$ 时， $\int_a^b y_m y_n \rho dx = 0$.

所以 $y_m(x)$ 与 $y_n(x)$ 加权正交.

证明下式在三种齐次边界条件下都为零

$$I = k(b)[y_n(b)y'_m(b) - y_m(b)y'_n(b)] - k(a)[y_n(a)y'_m(a) - y_m(a)y'_n(a)]$$

证明：令 I_1 = 上式第1项， I_2 = 上式第2项

(1)若为第一、二、三类边界条件，即 (以端点a为例)

$$\alpha y'_m(a) + \beta y_m(a) = 0, \alpha y'_n(a) + \beta y_n(a) = 0.$$

对 $\alpha = 0$ (第一类边界条件) 或 $\beta = 0$ (第二类边界条件),

显然 $I_2 = 0$, 同理 $I_1 = 0 \rightarrow I = 0$;

对第三类边界条件，由于 α 、 β 不同时为零，则

易得 $y_n(a)y'_m(a) - y_m(a)y'_n(a) = 0 \rightarrow I_2 = 0$, 同理 $I_1 = 0 \rightarrow I = 0$;

(2)若为有界条件，即 $k(a)=0$, $k(b)=0$, 从而 $I = 0$;

(3)若为周期条件，即 $k(a) = k(b)$, $y(a) = y(b)$, $y'(a) = y'(b)$,
显然 $I = 0$.

当 $m = n$ 时,

$$N_m^2 = \int_a^b |y_m(x)|^2 \rho(x) dx \rightarrow y_m(x) \text{ 的模方公式}$$

N_m : 本征函数 $y_m(x)$ 的模;

若 $N_m \equiv 1 \rightarrow y_m(x)$ 是归一化的;

正交归一化的本征函数可写成:

$$\int_a^b y_m(x) y_n(x) \rho(x) dx = \delta_{mn} = \begin{cases} 0 & m \neq n, \\ 1 & m = n. \end{cases}$$

性质4 (完备性) 本征函数系列 $y_n(x)$ ($n=1,2,3,\dots$) 是完备的, 即任一个有(分段)连续二阶导数和连续一阶导数的函数 $f(x)$, 可按此本征函数系列展开为一个绝对一致收敛的**广义傅里叶级数**

$$f(x) = \sum_{n=1}^{\infty} f_n y_n(x)$$

其中

$$f_n = \frac{\int_a^b y_n(x) f(x) \rho(x) dx}{\int_a^b |y_n(x)|^2 \rho(x) dx} = \frac{1}{N_n^2} \int_a^b y_n(x) f(x) \rho(x) dx$$

第2章 作业

教材P48-51:

1, 7, 8, 10, 14, 18, 23

第二章 分离变量法

- **熟练掌握：**

- 1、分离变量法的解题步骤，熟记齐次一维振动和热传导方程在第一、第二类边界条件下对应的特征值问题、特征值，特征函数系及形式解的结构；
- 2、圆域上Laplace方程边值问题的特征值问题、特征值、特征函数及形式解结构；
- 3、非齐次方程的解法；

- **掌握：**

- 1、非齐次边界条件的处理方法；
- 2、矩形域和扇域、环域上Laplace方程边值问题的特征值问题、特征值、特征函数及形式解结构；

- **理解：**

- 1、分离变量法的实质、适用范围；
- 2、定解问题的解的物理意义；
- 3、S-L方程本征值问题的一般提法；
- 4、S-L方程本征值问题的一般性质。